

Carisurg Week 1 | Sekou Ruddock

Project Title :Hybrid Clinical-Text Decision Support System (HCT-DSS) for Emergency Triage

Problem Statement

Triage nurses at Mercer General Hospital face high subjectivity, causing inconsistent over-triaging and under-triaging. In their paper-based, 12-bed Caribbean emergency department, which handles seasonal surges exceeding 150 patients without a dedicated pediatric unit, existing artificial intelligence frameworks offer inconsistent performance and lack validation for high-stakes clinical environments. Furthermore, live clinical outcome impacts remain completely unassessed. I propose a 12-week pilot of a supervised machine learning decision support tool integrating patient vitals and typed symptom notes with natural language processing to achieve at least a 85% Emergency Severity Index concordance against expert consensus and compile a queryable, de-identified digital database foundation.

Literature Review

As emergency departments worldwide face severe overcrowding and delays, researchers are increasingly evaluating artificial intelligence to streamline patient check-ins and predict clinical outcomes.

To address human errors caused by emergency room crowding, one study used a large clinical trial to compare an AI-supported check-in system directly against traditional manual frameworks. This smart tool successfully lowered patient mortality rates, boosted satisfaction, and optimized hospital resource use. However, the study was strictly limited by its single-center design and the exclusion of children, restricting its immediate use in broader healthcare systems. Building on the need to minimize human guesswork, a subsequent study trained six machine learning models using standard vital signs mixed with unstructured nurse notes. This natural language processing approach vastly outperformed real physicians at predicting critical hospitalizations and intensive care stays. Unfortunately, the model relied on a single data source, and its true clinical utility remains unproven since it has not yet been tested in a live environment. Shifting from classic machine learning to generative AI, another evaluation tested multiple Large Language Models to see how well they aligned with real doctor decisions. The models demonstrated promising but highly inconsistent triage performance, leading authors to recommend them only as supervised assistants rather than autonomous tools. Furthermore, this trial was constrained by its single-institution

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setting and simple text formatting errors that forced the exclusion of up to 9.6% of patient cases.

For time-sensitive, life-threatening emergencies like acute coronary syndrome, a systematic review of fifteen international studies examined advanced deep-learning frameworks. These automated models achieved exceptional diagnostic accuracy and significantly slashed critical treatment timelines like door-to-balloon intervals. Still, the overall findings were hindered by massive differences in study designs, varying data inputs, and a distinct lack of live, prospective validation. To overcome clinician hesitation surrounding hidden calculations, researchers trained highly interpretable frameworks using data from over 133,000 patient visits. The resulting model achieved over 91% accuracy while using explainable AI techniques to show medical staff exactly how vital signs and age drove triage decisions. However, a 2026 critical systematic review on explainable AI warns that such feature-attribution metrics frequently inflate clinician trust without improving decision quality, masking underlying model errors and worsening automation bias. Beyond pure math, underlying organizational workflows also dictate triage success, as shown by a qualitative study mapping patient traffic in a Caribbean hospital. This multi-method analysis established a core framework of strategies that either help or hinder patient transit through an under-represented healthcare system, though it was limited by its single-site design. Similarly addressing operational bottlenecks at early check-in, a streamlined math model used only basic front-door information like age and vitals to predict patient disposition. Nevertheless, its single-center retrospective design and strict exclusion of non-Asian encounters significantly limit its universal applicability.

Shifting focus to systemic deployment challenges, a broader systematic review evaluated the integration of machine learning and voice-based AI directly into clinical decision-support systems. While these tools optimized workflows and accelerated documentation speeds, they faced deep operational hurdles including data gaps, under-triage risks, and algorithmic opacity. The authors concluded that a heavy reliance on single-center data and a lack of standardized equity metrics continue to stall widespread adoption. This systemic equity deficit is starkly illustrated by the Optum care allocation algorithm failure, where optimizing for medical expenditures rather than objective clinical needs mathematically penalized and under-triaged sicker Black patients due to structural access barriers. Compounding this issue, a 2026 scoping review on triage ethics notes that uncalibrated monocentric models routinely replicate historical demographic wait-time disparities and cultivate a dangerous "automation dependency" among frontline staff. These practical roadblocks were mirrored in a qualitative study interviewing 19 Canadian healthcare experts about their real-world experiences rolling out AI check-in tools. The interviews brought to light major

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implementation issues, including outdated hospital computers, daily routine disruptions, and legal anxieties over software errors. Crucially, the findings revealed that these models are mostly optimized for wealthy hospitals, meaning they may create unfair care or fail entirely when dropped into understaffed clinics.

A scoping review of 47 studies echoed these concerns across both ambulance and hospital environments, noting that real-world rollouts face massive friction from low clinician trust and ethical worries. A central operational constraint identified was the hidden decision-making logic of these models, which operates as a "black box" to medical staff. This severe lack of transparency makes it incredibly difficult for doctors to confidently rely on the AI's math during high-stakes emergency scenarios. This operational friction is worsened by **emergency informed consent gaps**; because obtaining proactive, explicit consent is a structural impossibility during acute clinical crises, contemporary legal guides recommend operating under a framework of "presumed consent" paired with rigid post-encounter reconciliation loops. Finally, a simulation study evaluating a real-time text-processing alert tool for deadly heart conditions highlighted the immediate behavioral impacts of AI on clinicians. While the 14 participating doctors appreciated the clear, automated warnings, they raised urgent concerns regarding technical glitches and the psychological risk of automation bias leading to misdiagnoses. Ultimately, the tool's true clinical utility remains unproven due to a tiny participant pool and a heavy reliance on fake case simulations rather than live, high-pressure emergency environments.

Main two research gaps

- **Single-Center Reliance and Limited Generalizability:** The majority of the evaluated triage frameworks (including those analyzed by Boğa, Chang, and Nedos) heavily rely on retrospective datasets from a single institution. This lack of extensive external validation restricts the generalizability of their results to distinct clinical configurations, such as low-resource, stand-alone, or regional Caribbean facilities.
- **Deficit in Real-World, Live Clinical Outcome Assessments:** While the studies show promising mathematical performance or retrospective alignment, they consistently lack evaluation regarding direct, real-world prospective implementation. There is an unaddressed gap concerning how these tools dynamically affect patient workflow constraints and live clinical indicators (such as emergency department length of stay or mortality rates) when embedded in active clinical environments.

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Notes on the Mercer General ED triage process with AI implementation.

1. Patient Arrival & Registration

The patient arrives at the emergency department lobby and is logged in by an administrative clerk.

- **Information Entry Point:** Basic demographics, arrival time, and primary complaint are written on a physical paper intake chart.
- **Explicit Delay:** High-volume patient surges create an immediate physical queue at the front door before registration can even begin.

2. Vitals Capture

The triage nurse measures core physiological signs (blood pressure, heart rate, respiratory rate, oxygen saturation, and temperature).

- **Information Entry Point:** Vitals are manually written onto the patient's paper chart.
- **Explicit Delay:** Physically collecting a full set of vital signs takes a fixed amount of time per patient, causing a backlog in the waiting area.

3. Triage Nurse Assessment & Care Streaming

The nurse interviews the patient to evaluate their symptoms and assigns an urgency level (Emergency Severity Index).

- **AI Implementation:** The nurse types the collected vitals and brief symptom notes into a local desktop application. The AI processes this data in under 2 seconds, displaying a validated acuity recommendation and highlighting high-risk conditions.

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- **Decision Point:** The nurse decides on the patient's treatment stream and determines whether to start front-loaded investigations (such as immediate ECGs or point-of-care blood tests) right from the waiting area.
- **Where Information Gets Lost:** The central electronic database operates with a **5-to-30-minute system sync lag**. Because data entered at the triage desk is not instantly visible on the main department computers, physicians cannot see the digital triage findings or AI safety flags in real time. To prevent a communication blackout, staff must rely entirely on the physical paper chart traveling with the patient.

4. Main ED Assessment

The patient is moved into an acute bed or evaluation space when available, where the treating doctor or house officer performs a full clinical examination.

- **Handoff Point:** The physical paper chart is handed from the triage nurse to the treating physician.
- **Explicit Delay (Bed Blocking):** The department has a compact 12-bed capacity. When all beds are occupied by admitted patients waiting for an upstairs ward, a physical bottleneck occurs, stalling the entire flow.
- **Where Information Gets Lost:** If a patient's condition deteriorates while they are stuck waiting in the lobby due to bed shortages, that critical change is lost to the system unless a staff member manually performs a full re-triage.

5. Disposition & Patient Handover

After reviewing the patient's clinical response and test results, the physician makes the final disposition decision: either safely discharging them home with a follow-up plan or admitting them to the hospital.

- **Handoff Point:** Admitted patients are moved to a temporary ED holding bay or transferred to an inpatient ward team.
- **Where Information Gets Lost:** These internal transitions and staff shift changes rely heavily on verbal communication and loose paper notes. In a crowded, noisy environment, crucial clinical details such as subtle trends in vital signs, pain responses, or specific nursing concerns frequently drop out of the conversation during the handover.

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Proposed Solution

1. Digital Intake & Data Ingestion Layer

To eliminate the paper-based vulnerability and capture the necessary inputs for the machine learning model, a lightweight, locally hosted digital intake web application will be deployed at the triage desk.

Structured Data Input: Fields mapped directly to the Mercer Triage Record (Age, Temp, HR, RR, BP, SpO2, Blood Glucose, and Pain Score).

Pediatric & Obstetric Logic: Because there is no separate pediatric unit, the system will apply age-segmented physiological alert thresholds (e.g., flagging a respiratory rate of 30 as abnormal for an adult but normal for an infant).

Unstructured Text Capture: A dedicated free-text field for the nurse to type the "Chief Complaint & History" notes.

2. Dual-Stream Machine Learning Engine

The software architecture uses a supervised ensemble learning approach to process two distinct data streams simultaneously to predict the true ESI level (1 through 5):

The Vital Sign Stream (Numerical): A supervised classification model (such as a Gradient Boosting or Random Forest Classifier) evaluates the physiological variables against clinical danger zones.

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The Text Stream (Linguistic): A natural language processing (NLP) module processes the typed nurse notes. It tokenizes text, removes filler words, and extracts clinical risk keywords (e.g., identifying linguistic markers like "crushing chest pain" or "acute onset weakness").

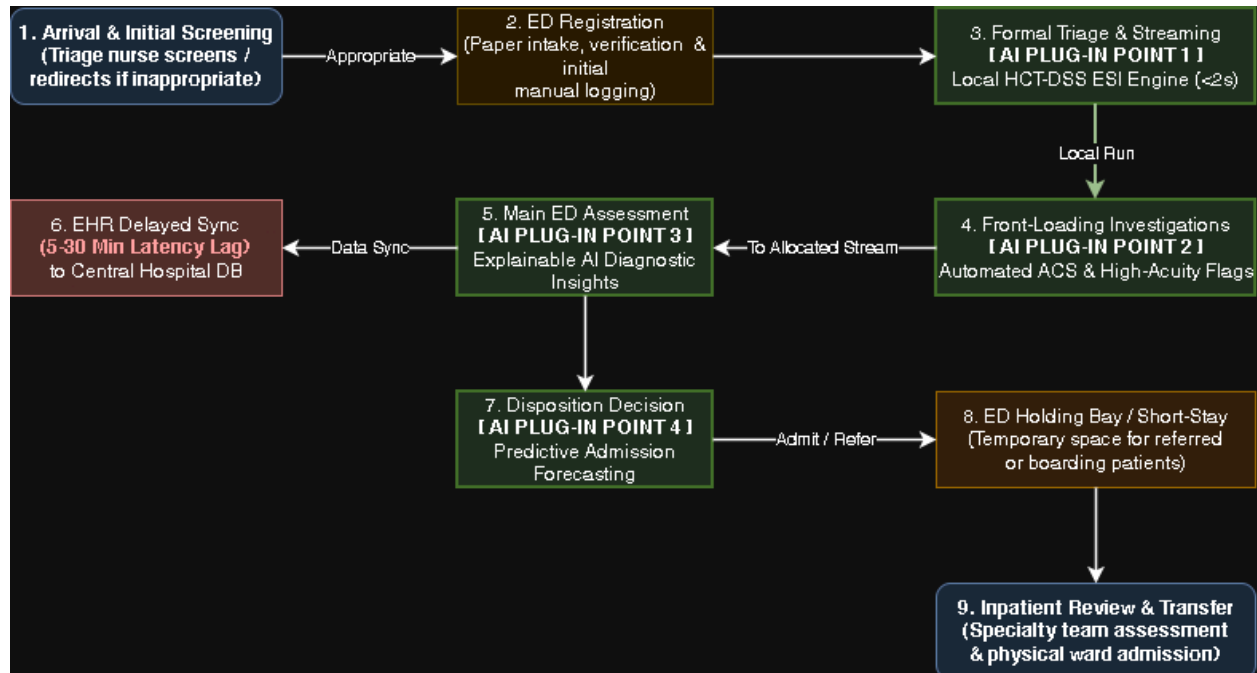
Ensemble Fusion: The engine combines the outputs of both streams, calculating a probability distribution across ESI levels and displaying the recommended level to the triage nurse within < 2 seconds of form submission.

3. Secure Logging and Human-in-the-Loop Validation

Upon form submission, an automated pipeline immediately strips all personal identifiers (such as names, addresses, and exact arrival times), writing purely anonymous clinical data to a secure local SQL database. The system functions strictly as a decision support tool; triage nurses retain final authority and log a reason code for any overrides. To validate accuracy, a weekly clinical consensus panel audits a randomized sample of entries and 100% of nurse overrides. This panel establishes the clinical ground truth to calculate and ensure the project reaches its target of at least 85% ESI concordance by Week 12.

Flow Diagram Showcasing how the HCT-DSS (Hybrid Clinical-Text Decision Support System) can be integrated inside of the Mercer General ED triage process.

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Workflow Constraints

1. Technical Constraint

- **Constraint Description:** Limited and intermittent external WAN/internet connectivity within the regional facility.
- **Category:** Technical Infrastructure.
- **Why It Causes System Failure/Harm If Ignored:** If the risk-stratification models rely on cloud-hosted external APIs (e.g., remote OpenAI endpoints), a network drop will cause the triage interface to freeze, or time out. In an intense ED environment, this delay forces the nurse to bypass the digital terminal entirely, breaking the data collection pipeline and leaving high-acuity patients without standardized decision support.

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- **Design Impact:** The HCT-DSS must be built as a fully self-contained, lightweight deployment hosted on a local Docker container running within the hospital's Local Area Network (LAN). The model must achieve 100% functionality fully offline.

2. Operational Constraint

- **Constraint Description:** High cognitive overhead and rapid throughput demands during seasonal tourist surges (over 150 patients daily), where a single nurse manages 15–30 high-acuity patients per shift.
- **Category:** Operational Capacity / Human Factors.
- **Why It Causes System Failure/Harm If Ignored:** Introducing multi-tiered menu selections, mandatory typing fields for validation, or high latency alerts disrupts the fluid workflow of a busy shift. Triage nurses under acute stress will experience "alert fatigue" and perceive the tool as an administrative hindrance rather than a clinical aid, resulting in immediate abandonment of the software.
- **Design Impact:** The User Interface must feature a single-screen dashboard layout that presents the suggested ESI score inline without overlay pop-ups. If the clinician disagrees with the model, an easy "one-click override" drop-down menu with standardized clinical reason codes must be provided to keep interaction times under 3 seconds.

3. Structural Constraint

- **Constraint Description:** Lack of physical space separation and specialized sub-units, meaning pediatric presentations and obstetric emergencies are managed within the same 12 general acute cubicles and 2 resuscitation bays.
- **Category:** Structural / Environmental.
- **Why It Causes System Failure/Harm If Ignored:** Models built on standard adult emergency datasets will miscalculate risks when encountering pediatric vitals (e.g., higher baseline respiratory and heart rates) or maternal physiological changes, causing dangerous under-triaging of vulnerable populations sharing the physical ward.
- **Design Impact:** The data entry interface must feature a mandatory, front-loaded demographic toggle ("Adult", "Pediatric", "Obstetric") that alters the underlying feature weights and references distinct alert thresholds optimized for specific physiological baselines.

Key Stakeholders

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Stakeholder Name & Role	Workflow Touchpoint	Primary Specific Concern	Operational Success	System Design Response
Sister Patrice Senior Triage Nurse	Operates triage desk application during patient intake.	Software clicks or modal pop-ups slowing down rapid sorting during volume surges.	Completing assessments fast with zero workflow interruptions.	Passive Sidebar: Dynamic inline tier display; no blocking dialogs or modal prompts.
Dr. Loren De Freitas ED Clinical Lead	Reviews patient charts and handles acute assessments.	Missing high-risk, atypical cases due to an uninterpretable "black-box" model.	Instantly verifying and auditing the clinical logic behind any AI score.	Explainable AI: Visual highlights over key vitals and symptom text driving the score.
Mr. Ricardo Clarke Chief IT Officer	Manages department servers and network infrastructure.	System crashes or downtime caused by unstable external WAN internet drops.	100% local software availability completely offline.	On-Prem LAN Deployment: Self-contained Docker container running on a local server.
Patient Advocate Ministry of Health	Monitors waiting room times and healthcare equity.	Algorithmic bias under-triaging patients who use regional clinical phrasings.	Equal triage accuracy across all local demographics and idioms.	Localized Training: NLP engine fine-tuned on historical, de-identified local ED charts.

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Hospital Administrator	Manages overall department throughput and bed capacity.	Over-triaging flooding the 12 acute beds and causing holding bay blocks.	Optimized patient flow hitting high agreement with expert clinical consensus.	Consensus Calibration: Model thresholds directly tuned against multi-expert datasets.
Chief Executive Officer				

Risk Analysis

Risk Name	Category	Likelihood (L/M/H) & Clinical Justification	Impact (L/M/H) & Clinical Consequence	Specific Mitigation (Who, When, Trigger)	Measurable Signal of Success

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Single-Center Training Overfitting & Generalizability Deficit	AI-Technical	High: The majority of existing models (Boğa, Chang, Nedos) rely on retrospective data from a single, wealthy institution that does not reflect a stand-alone Caribbean facility.	High: The model miscalculates triage levels when deployed in the local Mercer ecosystem, misallocating the limited 12 acute treatment beds.	Lead Data Scientist will validate the ensemble model against a localized, de-identified Mercer General historical dataset of 18 months before go-live. A drop of >3 percentage points in baseline recall triggers an immediate code deployment freeze.	External validation cohort agreement stays within ± 3 percentage points of training-site performance before pilot launch.
Formatting Ingestion Error & Patient Data Exclusion	AI-Technical	Medium: Literature shows text formatting inconsistencies in clinical environments can force the exclusion of up to 9.6% of patient triage cases.	High: High-acuity patients with typos or irregular free-text character inputs fail to generate a prompt within the target 2 seconds, stalling emergency streaming.	Lead Software Engineer will implement a regex-based automated input normalization layer in the digital intake app pre-deployment. Any unparsed text automatically triggers a silent fallback to a basic clinical keyword matching array.	Local system logs demonstrate a text-input rejection or failure rate of <1% during weekly system integrity checks.

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Intermittent WAN Infrastructure Disruption	Operational	High: Mercer General experiences documented, intermittent external WAN and internet connectivity drops within the regional facility.	High: Relying on external cloud APIs freezes the intake terminal, forcing nurses to abandon the digital data collection pipeline during heavy rushes.	Mr. Ricardo Clarke (IT Lead) will deploy the HCT-DSS container entirely on a local server within the hospital's physical Local Area Network (LAN). A WAN drop triggers an automatic routing to the local offline cache.	100% system availability and offline functionality verified during quarterly planned network infrastructure downtime drills.
Multi-System Ingestion Sync Asynchrony & Data Delays	Operational / Technical	High: Integrating manual front-desk digital intake with automated data streaming from legacy EHR systems or peripheral vitals monitors creates API data bottlenecks during peak surges.	High: Delays in combining manual inputs with automated data capture freeze the triage generation timeline, stalling clinical throughput and risking unmonitored patient deterioration in the waiting area.	Lead Software Engineer will build an asynchronous data pipeline using a local message broker queue. The UI will instantly display manually entered fields while background threads ingest and reconcile machine-captured data. Manual data pipelines are structurally prioritized over automated peripheral polling.	End-to-end data reconciliation latency between manual inputs and automated system capture remains under 1.5 seconds for 99% of active triage sessions.

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Surge-Driven Alert Fatigue & Software Abandonment	Operational	High: Triage nurses manage 15–30 high-acuity patients per shift during seasonal tourist surges exceeding 150 daily presentations.	Medium: Complex user interface choices or mandatory validation fields cause staff to experience alert fatigue and completely bypass the software.	The UI Designer will build a single-screen dashboard with a passive inline sidebar and a 1-click dropdown for override codes. If staff interaction time exceeds 3 seconds per patient, a workflow re-engineering review is triggered.	Application telemetry data monitors and confirms a clinician software utilization rate of $\geq 90\%$ across all active shifts weekly.
Automation Bias & Clinical Judgment Erosion	Ethical	Medium: Clinical simulation studies show a high psychological risk of doctors and nurses systematically deferring to automated warnings.	High: Clinicians accept an incorrect low-acuity score for an atypical presentation, leading to unmonitored patient deterioration in the waiting area.	Dr. Loren De Freitas (ED Clinical Lead) will mandate an interactive 1-hour automation bias training block pre-pilot. Clinician override rates dropping below 5% on complex cases triggers an immediate peer case review.	Triage staff correctly override incorrect model prompts in structured, blinded clinical simulation exercises at least 90% of the time.

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Emergency Informed Consent Gaps & Historical Data Blocks	Ethical / Legal	High: In acute emergency settings, seeking proactive informed consent to access deep historical data is often impossible due to patient alteration, unconsciousness, or urgent time constraints.	High: Missing critical historical records means the decision support system cannot scan for deadly drug allergies or underlying blood conditions (e.g., hemophilia), risking catastrophic, preventable adverse events.	Dr. Loren De Freitas (ED Clinical Lead) will oversee the integration of an audited "Emergency Break-the-Glass" consent override function within the HCT-DSS. Activating the override instantly unlocks critical allergy and hematological history while requiring a 1-click clinical justification, which automatically cues a post-encounter consent reconciliation workflow.	100% of unconsented historical data accesses are logged with valid clinical rationale, and administrative consent reconciliation is achieved within 24 hours of patient stabilization.
Accountability Ambiguity & Software Legal Anxieties	Ethical	Medium: Expert interviews reveal deep legal anxieties regarding software errors and ambiguous clinical fault lines when an AI tool is involved.	High: Institutional legal standards and a breakdown of medical trust among the local Mercer ED workforce if an adverse event occurs.	The Hospital Board will execute a formal operational SOP reinforcing absolute clinician autonomy over the tool. Dr. De Freitas is designated as the clinical owner responsible for auditing 100% of overrides weekly.	Executed accountability SOP is archived on permanent file, and 100% of weekly override audit logs are signed off on schedule.

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Shared Bed Asset Allocation & Pediatric Miscalculation	Equity	High: Existing research routinely excludes children, yet Mercer has no separate pediatric unit, managing kids within the 12 general beds.	High: Adult-trained models misinterpret elevated baseline pediatric vitals as high adult emergencies, or vice versa, causing dangerous under-triaging.	The Engineering Team implements a mandatory, front-loaded demographic toggle ("Adult/Pediatric/Obstetric") on the application. Omitting this selection blocks form submission and shifts physiological alert thresholds.	Disaggregated quarterly audits confirm that pediatric ESI concordance reaches the target metric of $\geq 85\%$ and matches adult accuracy within ± 3 percentage points.
Linguistic Idiom Misinterpretation & Demographic Bias	Equity	High: Standard natural language processing models fail to recognize regional Caribbean phrasing, idioms, or distinct nurse notation styles.	High: Local walk-in populations describing severe symptoms in regional dialects are systematically under-triaged, widening health disparities.	The ML Developer and local Patient Advocate Group will conduct a pre-pilot localized NLP embedding validation using historical Mercer charts. A subgroup recall gap of >5 percentage points triggers a linguistic retraining loop.	Bi-weekly algorithmic equity audits demonstrate that triage accuracy variance remains below 5 percentage points across all local dialect representations.

Root Cause Analysis: Optum Care Allocation Algorithm

A commercial health risk-prediction algorithm, widely utilized across the United States to target patients for intensive population health management programs, generated systemic racial disparities in its triage outputs. At identical risk percentiles, sicker Black patients were assigned scores equivalent to healthier White patients. For instance, at the 97th percentile auto-enrollment threshold, Black patients suffered from 26.3% more active chronic conditions than their White counterparts. This under-triage restricted access to crucial care resources; correcting this predictive gap would have more than doubled Black enrollment in the program, increasing it from 17.7% to 46.5%.

The core technical failure stems from flawed problem formulation and label choice. The algorithm did not train on an objective metric of health status; instead, it optimized for total medical expenditures (future healthcare costs) as a convenient proxy for health needs. Because sicker Black patients historically generate \$1,801 less in annual medical costs than equally sick White patients due to systemic access barriers, a model optimized to predict spending mathematically penalized Black individuals. The model was technically well-calibrated for dollar amounts but fundamentally misaligned with patient health.

The algorithm's optimization logic failed to anticipate data bias driven by structural inequality. It treated healthcare expenditure as a neutral, uncorrupted indicator of disease severity, ignoring the structural factors such as income, geography, and lower clinical utilization that create a wedge between a patient's true medical need and their financial cost footprint. Evaluating model calibration against multi-dimensional clinical endpoints during the validation phase would have caught the error. Testing model outputs against objective physiological markers (such as HbA1c for diabetes, systolic blood pressure for hypertension, and creatinine for renal failure) rather than purely financial records would have exposed the health disparity instantly.

System developers and healthcare administrators fell prey to an unverified behavioral assumption: that healthcare costs are an accurate, neutral proxy for how much medical help they require. Decision-makers prioritized the convenience of easily accessible insurance claims data over clinically grounded training datasets. Administrators failed to anticipate that human oversight protocols would be undermined by automation bias. While the clinical workflow allowed primary care physicians to manually review and refer to lower-scoring patients (between the 55th and 97th percentiles), clinicians only corrected a minimal fraction of the bias. Enrolled cohorts remained only 19.2% Black, failing to match the 26.9% representation that would have occurred if clinicians had actively targeted the sickest individuals within those risk pools. This demonstrates that humans over-relied on the algorithm's skewed screening thresholds.

A structured administrative audit comparing triage enrollment cohorts against actual localized disease registries would have intercepted this human-factors failure. Shifting the human framework from a cost-control mindset to a physiological-need mindset would have prompted an earlier review of the entry pipeline. Ultimately, changing the procedure to feed the algorithm an index variable combining health and cost predictions would have corrected the human design oversight, slashing the clinical bias by 84%.

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MEMORANDUM

TO: Dr. Loren De Freitas (ED Clinical Lead) and the Ministry of Health

FROM: Sekou Ruddock

DATE: June 30, 2026

SUBJECT: Risk Analysis for HCT-DSS Project Deployment

Verdict

The proposed system carries manageable risk if three conditions are met.

Top 3 Risks

- Risk 1 (Single-Center Training Overfitting): The software may make incorrect triage decisions because it was built using data from a single, wealthy overseas hospital that does not match the patient mix and limited resource constraints of our local facility.
- Risk 2 (Intermittent WAN Infrastructure Disruption): Frequent local internet dropouts could freeze the digital registration screens completely, forcing nurses to abandon the system during busy patient surges.
- Risk 3 (Linguistic Idiom Misinterpretation): The language processing algorithm may fail to understand local Caribbean phrases and regional dialects used by walk-in patients to describe serious symptoms, leading to them being mistakenly sent to the back of the line.

Mitigations

- Mitigation 1: The Lead Data Scientist will validate the model against an 18-month historical dataset from Mercer General before go-live, triggering an immediate freeze on code deployment if baseline model recall (accuracy in finding true positives) drops by more than 3 percentage points.
- Mitigation 2: Mr. Ricardo Clarke (IT Lead) will host the software container entirely on a local server within the hospital's physical Local Area Network (LAN), allowing the system to automatically route to a local offline cache when the internet drops.
- Mitigation 3: The Machine Learning Developer and a local Patient Advocate Group will test the language model using historical local patient charts, triggering a training loop to teach the system local phrasing if a sub-group accuracy gap exceeds 5 percentage points.

Signals of Success

- Signal 1: External validation testing confirms that the system's accuracy on local historical data stays within 3 percentage points of its original training performance before the pilot launch.

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- Signal 2: Technical logs verify 100% software availability and continuous offline functionality during quarterly planned network infrastructure downtime drills.
- Signal 3: Bi-weekly equity audits demonstrate that triage accuracy variance remains below 5 percentage points across all local dialect representations.

Residual Risk

At this stage, we cannot completely eliminate the risk of future, unmapped shifts in regional slang or catastrophic regional weather events that could physically destroy the local server infrastructure beyond our immediate hardware backup capacity.

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